Элементарная математика

Формулы сокращенного умножения

$$(a \pm b)^2 = a^2 \pm 2ab + b^2$$

$$(a \pm b)^3 = a^3 \pm 3a^2b + 3ab^2 \pm b^3$$

$$a^2 - b^2 = (a - b)(a + b)$$

$$a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2$$

Свойства степени

$$a^n b^n = (ab)^n \qquad (a^m)^n = a^{mn}$$

$$\frac{a^n}{b^n} = \left(\frac{a}{b}\right)^n \qquad a^0 = 1$$

$$a^m a^n = a^{m+n} \qquad a^{-n} = \frac{1}{a^n}$$

$$a^n \div a^m = a^{n-m} \qquad \sqrt[n]{a^m} = a^{\frac{m}{n}}$$

Свойства логарифмов

$$a^{\log_a b} = b \qquad \log_a a^b = b$$

$$\log_a b^c = c \log_a b \qquad \log_a bc = \log_a b + \log_a c$$

$$\log_a \frac{b}{c} = \log_a b - \log_a c \qquad \log_a b = \frac{\log_c b}{\log_c a}$$

$$\log_{a^c} b = \frac{1}{c} \log_a b$$

Формулы тригонометрии

$$\sin^2\alpha + \cos^2\beta = 1$$

$$\operatorname{tg}\alpha = \frac{\sin\alpha}{\cos\alpha} \qquad \operatorname{ctg}\alpha = \frac{\cos\alpha}{\sin\alpha}$$

$$\sec\alpha = \frac{1}{\cos\alpha} \qquad \operatorname{cosec}\alpha = \frac{1}{\sin\alpha}$$

$$1 + \operatorname{tg}^2\alpha = \frac{1}{\cos^2\alpha} \qquad 1 + \operatorname{ctg}^2\alpha = \frac{1}{\sin^2\alpha}$$

$$\sin 2\alpha = 2\sin\alpha\cos\alpha$$

$$\cos 2\alpha = \cos^2\alpha - \sin^2\alpha$$

$$\operatorname{tg} 2\alpha = \frac{2\operatorname{tg}\alpha}{1 - \operatorname{tg}^2\alpha}$$

$$\sin^2\frac{\alpha}{2} = \frac{1-\cos\alpha}{2} \qquad \cos^2\frac{\alpha}{2} = \frac{1+\cos\alpha}{2}$$

$$\operatorname{tg}\frac{\alpha}{2} = \frac{1-\cos\alpha}{\sin\alpha} = \frac{\sin\alpha}{1+\cos\alpha}$$

$$\sin\alpha = \frac{2tg\frac{\alpha}{2}}{1 + \operatorname{tg}^2\frac{\alpha}{2}} \qquad \cos\alpha = \frac{1 - \operatorname{tg}^2\frac{\alpha}{2}}{1 + tg^2\frac{\alpha}{2}}$$

$$\operatorname{tg}\alpha = \frac{2\operatorname{tg}\frac{\alpha}{2}}{1 + \operatorname{tg}^2\frac{\alpha}{2}} \qquad \operatorname{ctg}\alpha = \frac{1 - \operatorname{tg}^2\frac{\alpha}{2}}{2\operatorname{tg}\frac{\alpha}{2}}$$

$$\sin\alpha\cos\beta = \frac{1}{2}(\sin(\alpha+\beta) + \sin(\alpha-\beta))$$

$$\cos\alpha\cos\beta = \frac{1}{2}(\cos(\alpha+\beta) + \cos(\alpha-\beta))$$

$$\sin\alpha\sin\beta = \frac{1}{2}(\cos(\alpha-\beta) - \cos(\alpha+\beta))$$